SACHIN ANANDHARAJA

B.TECH - Artificial Intelligence And Data Science

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EXECUTIVE SUMMARY

B.Tech Artificial Intelligence and Data Science with hands-on experience in robotics, embedded systems, computer vision, ESP32-based projects, and data analysis. Experienced in developing practical hardware-software systems using C/C++, Python, ESP32, OpenCV, ROS 2, Gazebo, and FreeCAD. Interested in robotics software, autonomous systems, embedded robotics, robot simulation, and real-time hardware integration. Strong focus on learning through practical projects and building deployable robotics systems.

EDUCATION

Gnanamani College of Engineering & Technology ,Namakkal

B.Tech(AI&DS) , 2023-2027 | CGPA: 7.3/10

Don Bosco Higher Secondary school,Polur

HSC ,2022-2023

PROJECTS

ESP32 Pick-and-Place Robotic Arm

Technologies: ESP32, C/C++, MG996R, SG90, Servo Motors, FreeCAD

- Designed and developed a multi-servo robotic arm prototype for pick-and-place operations.
- Integrated ESP32 with multiple servo motors for controlling robotic joints.
- Worked on servo control, PWM-based positioning, power requirements, and hardware testing.
- Designed the mechanical model using FreeCAD with the goal of creating a digital representation of the physical robot.
- Planned future integration with ROS 2 and Gazebo for simulation and autonomous control.

ESP32 Wi-Fi Controlled Robot Car

Technologies: ESP32, C/C++, Blynk IoT, L298N, DC Motors, IR Sensor

- Developed a Wi-Fi-controlled mobile robot using ESP32 and Blynk IoT.
- Controlled DC motors through an L298N motor driver.
- Integrated sensors for monitoring the robot's operating conditions.
- Implemented real-time hardware control through a wireless interface.
- Explored intelligent safety features using sensor-based decision making.

ACHIEVEMENTS

Smart India Hackathon (SIH) – Inter-College Hackathon

Selected among participating teams at the inter-college level for presenting an innovative Electronics Queue Status Indicator solution, demonstrating skills in problem-solving, electronics, teamwork, and project development.

TECHNICAL PROFICIENCIES

Technical Skills

Programming

- Python
- C/C++

CAD & Simulation

- FreeCAD
- TinkerCAD
- Gazebo

ROBOTICS & AUTOMATION

- ROS 2
- Linux
- RViz
- Robot Kinematics
- URDF

Professional Competencies

- Team work
- Problem Solving
- Leadership
- Communication

Declaration:

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief